

Heat-Gated Fourier Features

Coordinate MLPs tend to learn low-frequency spatial structure more easily than sharp variations. Fourier features provide the network with an explicit high-frequency representation and have been shown to substantially improve the ability of MLPs to fit high-frequency functions in low-dimensional coordinate domains [1]. Our checkerboard is a particularly simple example: its density has sharp, axis-aligned boundaries, so relatively high spatial frequencies are needed to represent its support accurately. Unlike the random Fourier features of [1], we exploit the known geometry of the problem and use deterministic frequencies along each coordinate axis, log-spaced between $\omega_{\min}$ and $\omega_{\max}$. Log spacing gives coverage of coarse and fine spatial scales with relatively few features. The additional difficulty for diffusion and flow models is that the same network is used across an entire probability path. Near the data distribution, high-frequency features are useful for resolving sharp boundaries; at noisier timesteps, Gaussian smoothing has removed much of this fine structure. Heat-Gated Fourier Features (HGFF) therefore attenuate each Fourier feature according to the smoothing scale at the current timestep.

Derivations Consider a Gaussian corruption path

$$X_t = a_t X_{\text{data}} + s_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

Its Fourier transform is

$$\hat{p}_t(\omega) = \hat{p}_{\text{data}}(a_t \omega) \exp\left(-\frac{1}{2} s_t^2 \|\omega\|^2\right).$$

Thus Gaussian smoothing attenuates a spatial frequency ω by the factor

$$h(\omega, t) = \exp\left(-\frac{1}{2} s_t^2 \|\omega\|^2\right),$$

with high frequencies suppressed more rapidly than low frequencies. We choose K deterministic log-spaced frequencies

$$\omega_k = \omega_{\min} \left(\frac{\omega_{\max}}{\omega_{\min}}\right)^{\frac{k-1}{K-1}}, \quad k = 1, \dots, K,$$

and apply them independently to each coordinate dimension. The resulting feature is

$$\gamma_{d,k}(x, t) = \exp\left(-\frac{1}{2} s_t^2 \omega_k^2\right) \cos(\omega_k x_d + b_{d,k}),$$

where $b_{d,k}$ is a fixed phase. Concatenating over dimensions and frequencies gives

$$\gamma(x, t) = \bigoplus_{d=1}^D \bigoplus_{k=1}^K \gamma_{d,k}(x, t).$$

The construction therefore combines the Fourier-feature idea of exposing high-frequency spatial structure to a coordinate MLP [1] with two problem-specific choices: axis-aligned, log-spaced frequencies for the checkerboard, and a time-dependent gate derived from the exact attenuation of Fourier modes under Gaussian smoothing.

References

- [1] M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng. Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains. *NeurIPS*, 2020.